Siddhant Singhania
Electrical Engineering
Indian Institute of Technology Bombay

24B3956
Dual Degree (B.Tech. + M.Tech.)
Gender: Male
DOB: 04/04/2006

Examination	University	Institute	Year	CPI / % Credits
Graduation	IIT Bombay	IIT Bombay	2029	9.24 **Core:** 136 **| Total:** 139
Intermediate	CBSE	Shiv Nadar School Noida	2024	98.40%
Matriculation	CBSE	Shiv Nadar School Noida	2022	99.20%

Pursuing a **Minor Degree** in **Artificial Intelligence and Data Science** offered by **C-MInDS, IIT Bombay**

SCHOLASTIC ACHIEVEMENTS

- Received **AA/AB Grade** in **11** out of **14 core courses** in second year, achieving 9.42 SPI in 4th semester *('25-'26)*
- Awarded **Academic Scholarships** at Shiv Nadar School Noida consecutively in the years 2022, 2023 and 2024

TECHNICAL EXPERIENCE

Digital Design Trainee | *Cadence Design Systems* *(May'26-Jul'26)*

Performed Static Timing Analysis and Chip Optimization on clients' chip designs using industry-grade tools

- Studied, explored and performed **Static Timing Analysis** on several digital design Verilogs using **Cadence Tempus**, evaluating delay metrics such as **setup-hold** violations**, on chip variations** and **path chip pessimism**
- Understood **propagation delay** more deeply, analysing **transition rates, load capacitances** across paths, **latencies,** and delay models used in interconnect **cell delay calculations** such as **NLDM** and **CSM models**
- Designed an end-to-end **MCP Client-Server** Python Application built on Cadence's AI Architecture to perform **fully automated chip design constraint validation** using Cadence's newly acquired **TimeVision Software**

AI Intern - Sales Automation Pipeline | *Shorthills AI* *(May'25–Jul'25)*

Created an AI-driven sales automation pipeline to increase sales responsiveness, efficiency and engagement

- Engineered a sales automation system utilizing **MCP Servers** and **Large Language Model (LLM)** agents, reducing the sales lead response times from several days **to under 30 seconds** while handling large client volume
- Integrated personalized interaction workflows to enhance lead engagement, improving overall **lead communication**
- Led the design of the algorithm and pipeline architecture, using **Apache NiFi** and **Python** for flow-based product execution with **low latency, high packet handling, and seamless integration** with multiple APIs and tools

Electrical Subsystem | *IIT Bombay Racing* *(Jan'25–Apr'26)*

*Collaborated with a team of **100+** students to build a high-performance Electric Race Car for the prestigious Formula Student Switzerland and Portugal student race car competitions, organized by the Institution of Mechanical Engineers*

- Designed an **extended Kalman Filter Model** on **Embedded C** to predict **the state-of-charge of a 2RC Cell Model** in real time, and obtained a **high accuracy** on it in comparison to real SOC values from the accumulator
- Executed **circular daisy chain** on our battery monitoring system, successfully monitoring **96 cells** simultaneously
- Implemented and worked with **communication protocols** like **ISO-SPI, SPI, and CAN** for system integration

KEY PROJECTS

Real-Time Sobel Edge Detection Accelerator on FPGA | *Research Project* *(May'26–Present)*

Guide: Prof. Debanjan Bhowmik

- Implemented a **Sobel edge-detection pipeline** in **Verilog** using **circular line buffers** for live **3×3 windowing**, with a **convolution engine** computing edge metric against a threshold for real-time **binary edge classification**
- Validated pixel accuracy against a **NumPy-based reference model** on **BMP images** using a custom testbench
- Implementing the design on **Xilinx Vivado** for the **PYNQ-Z2** board, laying the groundwork for **CNN on FPGA**

I2C Device Driver Development on 8051 Microcontroller | *Course Project* *(Mar'26–Apr'26)*

Microprocessors Laboratory | Guide: Prof. Sachin Patkar

- Wrote bare-metal firmware on an **8051-based PT-51** board to interface with a **BMP280** pressure sensor via I2C
- Developed a custom **I2C device driver** library in C, implementing the protocol-standard Start, Stop, Read, Write, Ack, and Nack routines to enable **generic master-slave communication** with **multi-purpose I2C devices**
- Interfaced a **DS1307 RTC** module using the driver to fetch, read, and decode BCD-formatted timestamps over the **I2C** protocol, displaying **live time and temperature** on the master LCD and streaming them via **UART**

Digital Logic Design using VHDL | *Course Project* *(Aug'25–Nov'25)*

Digital Circuits Laboratory | Guide: Prof. Saurabh Lodha

- Designed a **Run-Length Encoder-Decoder** pair in VHDL with a custom FSM to traverse an 8x8 input matrix
- Implemented a **1024-bit RLE buffer** for lossless compression and row-major reconstruction of the input matrix
- **Simulated, verified, synthesized, and deployed** the design on **FPGA hardware**, performing pin planning to map the 50 MHz clock, reset switch, pass/fail LEDs, and I/O constraints using the provided SDC timing constraints

WashU-2 CPU: 16-bit Accumulator-Based Processor | *Course Project* *(Mar'26–Apr'26)*

Microprocessors | Guide: Prof. V Rajbabu

- Studied and modified a **17-state, multi-cycle 16-bit CPU** in VHDL, refining its **fetch-decode-execute cycle**
- Enhanced the **instruction encoding** and **ALU design** across a diverse set of arithmetic and branching instructions
- Implemented **ALU-FSM logic** for the ADD and BRPOS (Branch-if-Positive) instructions, performing unsigned arithmetic, sign-bit/zero-flag condition checks, and authored comprehensive **VHDL testbenches** for verification

Phase Sensitive Detection (Lock-in Amplifier) | *Course Project* *(Mar'26–Apr'26)*

Analog Circuits Laboratory | Guide: Prof. Sandip Mondal

- Designed an op-amp based **all-pass phase shifter**, deriving its transfer function and verifying it experimentally
- Built a switch-based **Phase Sensitive Detector** using **CD4066** quad switches driven by complementary CMOS square-wave references, rectifying the input signal based on phase, on both **LTSpice simulation** and **hardware**
- Implemented a first-order active **low-pass filter** to extract the DC component from the PSD output, deriving its cut-off frequency and verifying the lock-in amplifier output across all tested phase angles, obtaining minimal error

Solving Coupled Differential Equations on FPGA | *Course Project* *(Aug'25–Nov'25)*

Digital Systems | Guide: Prof. Debanjan Bhowmik

- Designed **N parallel computational units** connected via **common bus architecture** to enable simultaneous equation solving and implemented **Euler integration method** in Verilog HDL for differential equation solutions
- Solved **Kuramoto oscillator differential equations** simultaneously through **FPGA hardware** to achieve significant computational speedup over traditional CPU computation through the parallel processing architecture
- Simulated and verified our **RTL designs**, yielding a $\mathbf{N^2/N}$ **speedup factor** compared to serial implementation

TECHNICAL SKILLS

Programming / Hardware	C | C++ | Python | VHDL | TCL | Verilog | 8051 Microcontroller | STM Microcontroller
Software	Quartus | MATLAB | LTSpice | Fusion360 | Tera Term | Tempus | Joules RTLStudio | Vivado

KEY COURSE

Computer Science	Computer Programming and Utilization, Programming for Data Science
Mathematics	Linear Algebra, Differential Equations, Calculus, Probability and Random Processes, Regression Analysis
Electrical	Power Engineering, Analog Circuits, Signal Processing, Digital Systems, Power Engineering-II, Microprocessors, Control Systems
Entrepreneurship	Introduction to Innovation & Entrepreneurship, Strategy and Leadership for Entrepreneurs

EXTRA-CURRICULAR ACTIVITIES

Leadership	• Selected as **Vice Captain** of the Career Guidance Cell at Shiv Nadar School, Noida, coordinating 20+ **career and college fairs** with top universities for **500+ students** *(2022)*
Technical	• Competed in **XLR8** organized by the Electronics & Robotics Club, developing a **Raspberry Pi Pico** based **joystick car** to navigate a complex obstacle course under fifteen minutes *(2024)* • Took part in **Winter in Data Science**, where I performed **RL** based modeling on **geopolitical disruptions** in global supply chains, using **multi-agent deep RL (MARL) framework** across **manufacturer-distributor-warehouse-retailer-customer echelons** *(2025)*
Basketball	• Completed two semesters of training in **Basketball** under National Sports Organization *(2024)* • Awarded **Bronze Medal** in the **IIT Bombay Freshiesta Basketball Tournament** *(2024)* • Secured **Fourth Position** in the **IIT Bombay Basketball General Championship** *(2025)* • Trained as one of **IITB's top 20 basketball athletes** in the **Inter IIT Pre-Camp** *(2025)*
International Exposure	• Represented my school as part of a 5-student team at the **Swiss Innovation Camp** in Geneva, Switzerland; attended advanced STEM workshops at **CERN** and other research centres *(2018)*